Project: Worked examples
Title: Fault protection (BS 7671)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates automatic disconnection of supply in a TN system for one circuit: loop impedance, disconnection time and the adiabatic conductor check. It also gives the prospective three-phase fault at the origin against the breaking capacity.

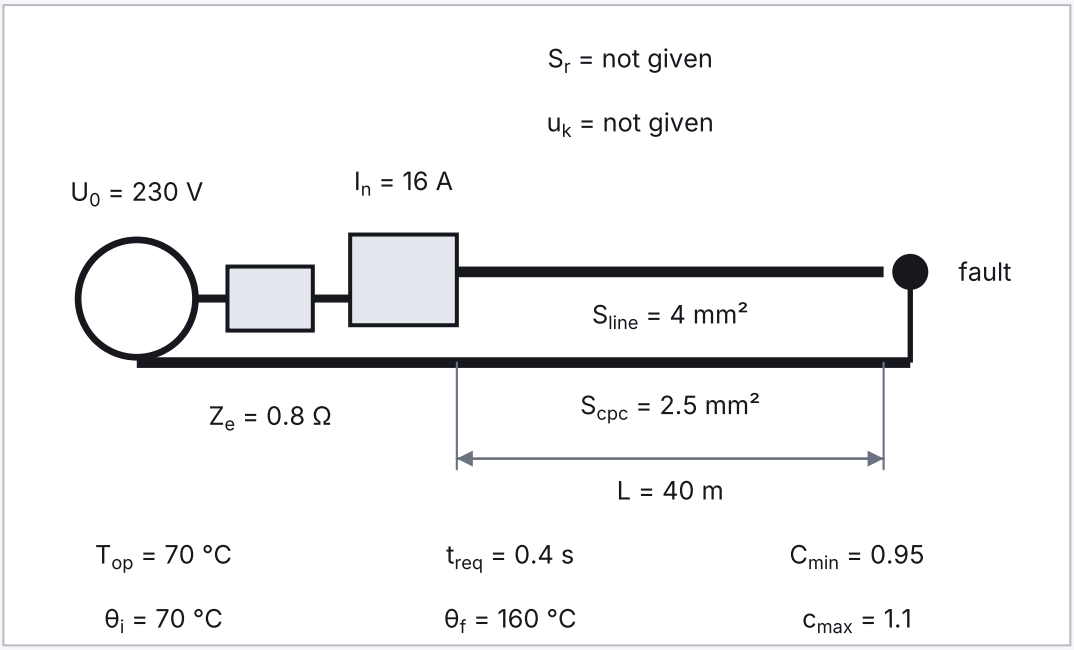


Figure 1 — The earth-fault loop of a circuit on a type C MCB: the supply U_0 behind Z_e, the device I_n, line and protective conductors of length L to the fault. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Loop impedance	$Z_s = Z_e + (R_1 + R_2)\cdot[1 + \alpha(\theta - 20)]$, $R = k_R\cdot\rho_{20}\cdot L/S$ ($k_R = 1.125$ takes the pure-copper ρ_{20} to the cable standard's maximum conductor resistance; k enters with the pure ρ_{20})
Fault current	$I_f = C_{min}\cdot U_0/Z_s$, $C_{min} = 0.95$
Maximum loop impedance	$Z_{s,max} = C_{min}\cdot U_0/I_a$; I_a = 5, 10 or 20·I_n for MCB types B, C, D, or read from the time–current points at t_{req} (log-log)
Adiabatic conductor check	$S_{min} = \sqrt{(I^2t)/k}$, $k = \sqrt{[Q_c(\beta + 20)/\rho_{20}\cdot \ln(1 + (\theta_f - \theta_i)/(\beta + \theta_i))]}$ (derived); $I^2t = I_f^2\cdot t_d$ from 0.1 s; below 0.1 s the device's let-through I^2t at the maximum prospective fault bounds both conductors at the origin (reg. 434.5.2, 543.1.3), and without it the sheet fails 'Conductors against let-through at the origin'

WHAT IT CALCULATES	RELATION
Protective device check	the device chosen here must be the one carried from the cable step (device$_{code}$ 1 = MCB B, 2 = C, 3 = D, 4 = fuse or MCCB), else the sheet is blocked
Fault level (simplified)	$Z_T = u_k \cdot U_n^2 / S_r$, $I_{k3} = c \cdot U_n / (\sqrt{3} \cdot Z_T)$, $U_n = \sqrt{3} \cdot U_0$

Basis: Automatic disconnection in TN systems: times and the loop impedance condition[1](reg. 411.3.2.2, 411.3.2.3, 411.4.4); Protective conductor and fault current adiabatic checks[1](reg. 543.1.3, 434.5.2); Protection by automatic disconnection of supply[2](cl. 6.3.3); Voltage factor c[3](cl. 5.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
device	Protective device	MCB type C (I_α = 10 I_n)		drawing: E-SCH-002 rev B	any
I_n	Rated current	16	A	drawing: E-SCH-002 rev B	6 … 630 A
L	Circuit length	40	m	drawing: E-LAY-002 rev B	1 … 500 m
S_{line}	Line conductor cross-section	4	mm²	drawing: E-SCH-002 rev B	1 … 300 mm²
S_{cpc}	Protective conductor cross-section	2.5	mm²	drawing: E-SCH-002 rev B	1 … 300 mm²
ρ_{20}	Resistivity at 20 °C (used for k, and for $R_1 + R_2$ through k_R)	0.000017241	Ω·mm	assumption: Annealed copper resistivity.	0.000015 … 0.00005 Ω·mm
k_R	Conductor resistance factor on ρ_{20} for $R_1 + R_2$ (1 = the resistivity itself; the default of 1.125 takes pure copper to the cable standard's maximum conductor resistances, so Z_s is not understated; k is not changed by it)	1.125		assumption: Cable standard maximum conductor resistance over the pure-copper resistivity (5 to 12 % higher); 1.125 covers the sizes used.	1 … 1.25
α	Temperature coefficient of resistance	0.00393	1/K	assumption: Copper temperature coefficient.	0.0035 … 0.0045 1/K
Q_c	Volumetric heat capacity	0.00345	J/(K·mm³)	assumption: Copper volumetric heat capacity.	0.002 … 0.005 J/(K·mm³)
β	Reciprocal temperature coefficient β	234.5	K	assumption: Copper reciprocal temperature coefficient.	200 … 260 K

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
θ_i	Initial temperature for the adiabatic check	70	°C	assumption: Thermoplastic operating temperature.	40 … 120 °C
θ_f	Final temperature for the adiabatic check	160	°C	assumption: Thermoplastic limiting temperature.	140 … 300 °C
U_0	Nominal voltage to earth	230	V	spec: Supply characteristics	100 … 600 V
Z_e	External earth fault loop impedance	0.8	Ω	spec: DNO declared maximum for TN-S	0.05 … 2 Ω
T_{op}	Conductor temperature for Z_s	70	°C	assumption: Thermoplastic insulation at full load.	30 … 90 °C
C_{min}	Voltage factor for the minimum fault	0.95		assumption: Minimum voltage factor.	0.93 … 1
c_{max}	Voltage factor c for the maximum fault	1.1		assumption: Maximum voltage factor.	1.05 … 1.1
t_{req}	Required disconnection time	0.4	s	spec: Final circuit up to 32 A, TN system	0.2 … 5 s
$I2t_{let}$	Device let-through energy I^2t at the maximum prospective fault (disconnection below 0.1 s)	1500	$A^2·s$	assumption: Read from the MCB let-through chart at the fault current.	100 … 50000000 $A^2·s$

3. Assumptions

A1. The system is TN with a single earth fault of negligible impedance. (Scope of the check.)
A2. There is no motor contribution, no arc fault and no IT system. (Scope of the check.)
A3. The line and protective conductors are the same material at the same temperature. (Conductor data from a material record.)
A4. The resistance is $\rho_{20}·L/S$ corrected by α. (Conductor data from a material record.)
A5. The factor k is derived from the material constants, not read from a licensed table. (Derived constants, not read from a table.)
A6. Fuse and MCCB characteristics are points the user reads off the manufacturer's curve. (No time–current table ships.)
A7. The points are interpolated log-log, with no extrapolation. (No time–current table ships.)
A8. The optional field-test value $0.8 \times Z_{s,max}$ is a rule of thumb for measurements at ambient temperature. (Labelled on the sheet.)
A9. The line conductor check (BS 7671 clause 434.5.2) needs the device's let-through energy at the maximum prospective fault. (Stated limitation: the end-of-line earth fault is the minimum fault, not the maximum.)
A10. The end-of-line earth fault is the minimum fault and does not bound that energy. (Stated limitation: the end-of-line earth fault is the minimum fault, not the maximum.)
A11. The line conductor check is made only when $I2t_{let}$ is entered or the disconnection time is 0.1 s or

more. (Stated limitation: the end-of-line earth fault is the minimum fault, not the maximum.)
A12. ρ_{20} (Resistivity at 20 °C (used for k, and for $R_1 + R_2$ through k_R)) is taken as 0.000017241 Ω·mm. (Annealed copper resistivity.)
A13. k_R (Conductor resistance factor on ρ_{20} for $R_1 + R_2$ (1 = the resistivity itself; the default of 1.125 takes pure copper to the cable standard's maximum conductor resistances, so Z_s is not understated; k is not changed by it)) is taken as 1.125. (Cable standard maximum conductor resistance over the pure-copper resistivity (5 to 12 % higher); 1.125 covers the sizes used.)
A14. α (Temperature coefficient of resistance) is taken as 0.00393 1/K. (Copper temperature coefficient.)
A15. Q_c (Volumetric heat capacity) is taken as 0.00345 J/(K·mm³). (Copper volumetric heat capacity.)
A16. β (Reciprocal temperature coefficient β) is taken as 234.5 K. (Copper reciprocal temperature coefficient.)
A17. θ_i (Initial temperature for the adiabatic check) is taken as 70 °C. (Thermoplastic operating temperature.)
A18. θ_f (Final temperature for the adiabatic check) is taken as 160 °C. (Thermoplastic limiting temperature.)
A19. T_{op} (Conductor temperature for Z_s) is taken as 70 °C. (Thermoplastic insulation at full load.)
A20. C_{min} (Voltage factor for the minimum fault) is taken as 0.95. (Minimum voltage factor.)
A21. c_{max} (Voltage factor c for the maximum fault) is taken as 1.1. (Maximum voltage factor.)
A22. $I2t_{let}$ (Device let-through energy I^2t at the maximum prospective fault (disconnection below 0.1 s)) is taken as 1500 A²·s. (Read from the MCB let-through chart at the fault current.)

4. Working

[19] $R_{12} = \rho_{20} * k_R * L * (1 / S_{line} + 1 / S_{cpc}) * (1 + \alpha * (T_{op} - 20\ degC))$
$= 0.000017241\ ohm·mm \times 1.125 \times 40\ m \times (1 / 4\ mm^2 + 1 / 2.5\ mm^2) \times (1 + 0.00393\ 1/K \times (70\ degC - 20\ degC))$
$= 0.60339\ ohm$
$R_1 + R_2$ at the conductor temperature

[20] $Z_s = Z_e + R_{12}$ [1](reg. 411.4.4)
$= 0.8\ ohm + 0.60339\ ohm$
$= 1.4034\ ohm$
Earth fault loop impedance

[21] $I_f = C_{min} * U_0 / Z_s$
$= 0.95 \times 230\ V / 1.4034\ ohm$
$= 155.69\ A$
Earth fault current

[22] $I_a = 10 * I_n$ [2](60898-1)
$= 10 \times 16\ A$
$= 160\ A$
Instantaneous tripping current (10 I_n)

[23] $Z_{s,max} = C_{min} * U_0 / I_a$ [1](reg. 411.4.4)
$= 0.95 \times 230\ V / 160\ A$
$= 1.3656\ ohm$
Maximum earth fault loop impedance

[24] $u_{Zs} = Z_s / Z_{s,max}$
$= 1.4034\ ohm / 1.3656\ ohm$
$= 1.0277$
Loop impedance utilisation

[25] $Z_{test} = 0.8 * Z_{s,max}$
$= 0.8 \times 1.3656\ ohm$
$= 1.0925\ ohm$
Field-test maximum at ambient (0.8 × $Z_{s,max}$, rule of thumb)

5. Results

SYMBOL	VALUE	DESCRIPTION
Z_s	1.4034 Ω	Loop impedance
I_f	155.69 A	Earth fault current
R_{12}	0.6034 Ω	$R_1 + R_2$
I_a	160 A	Disconnection current
$Z_{s,max}$	1.3656 Ω	Maximum loop impedance
Z_{test}	1.0925 Ω	Field-test maximum (rule of thumb)
u_{Zs}	1.03	Loop impedance utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Fault current against the disconnection current	I_f below the instantaneous tripping current	disconnection within t_{req}	–	[1](reg. 411.3.2.2)
FAIL	Loop impedance against the maximum	1.4034 Ω	1.3656 Ω	1.03	[1](reg. 411.4.4)

7. Applicability

WARNING records: tabulated values not taken from a reference record are ρ_{20}, k_R, α, Q_c, β, θ_i, θ_f (Conductor material constants); I^2t_{let} (Device characteristic (time–current points)). Enter each as one so its source, edition and licence travel with it. [record]
WARNING I_f: the device does not disconnect within the data given, so the adiabatic check cannot be made. Reduce the loop impedance or choose a faster device. [adiabatic]

8. Verification

Template elec.fault-protection@1.6.0 Content hash 627284e2d716cdc6 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 60204-1:2018.
3. IEC 60909-0:2016.